

Mechanics of Solids (3130608)



Dedicated Faculty, Committed Education

Darshan
Institute of Engineering & Technology

Centroid



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Syllabus – Module 3 Description



GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering

Subject Code: 3130608

Semester III

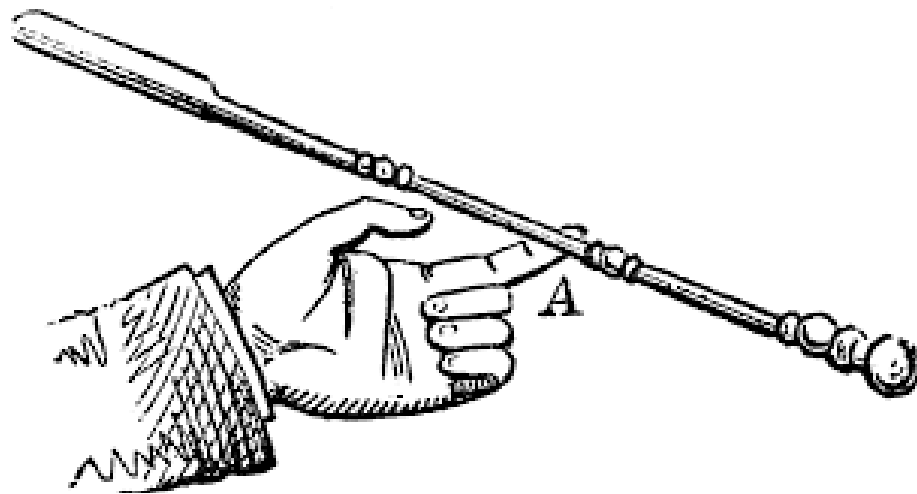
Mechanics of Solids

MODULE 3

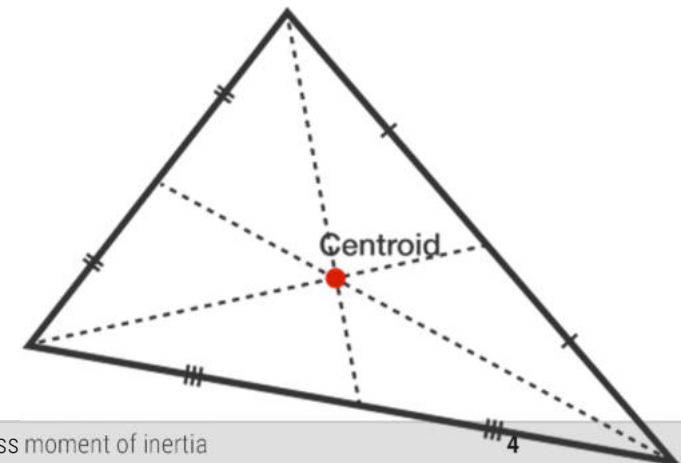
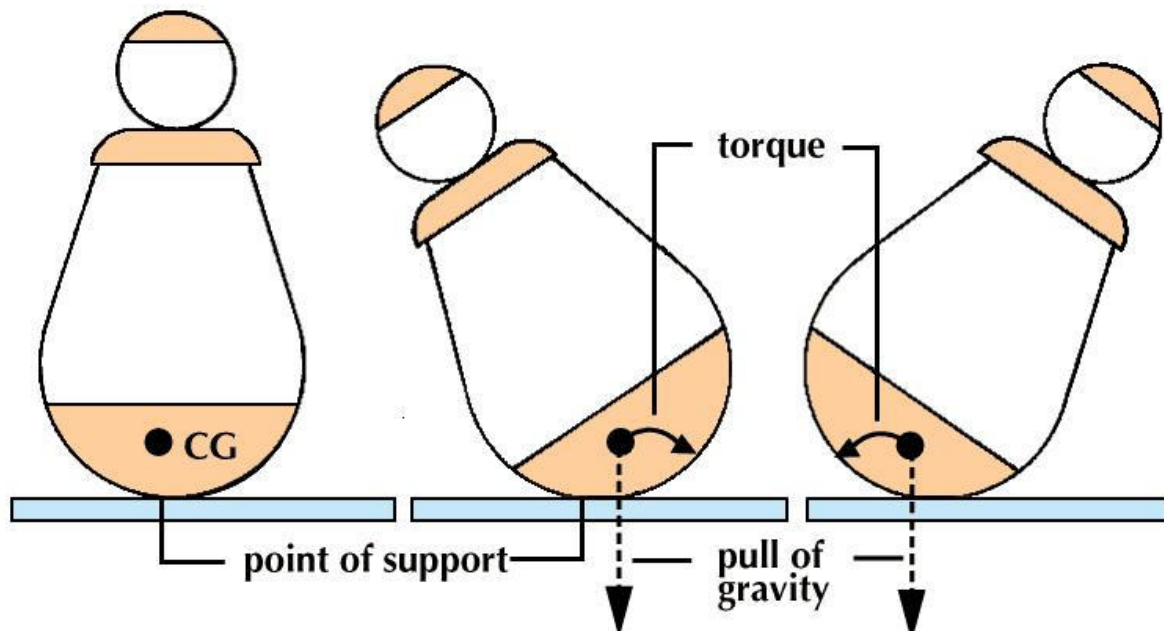
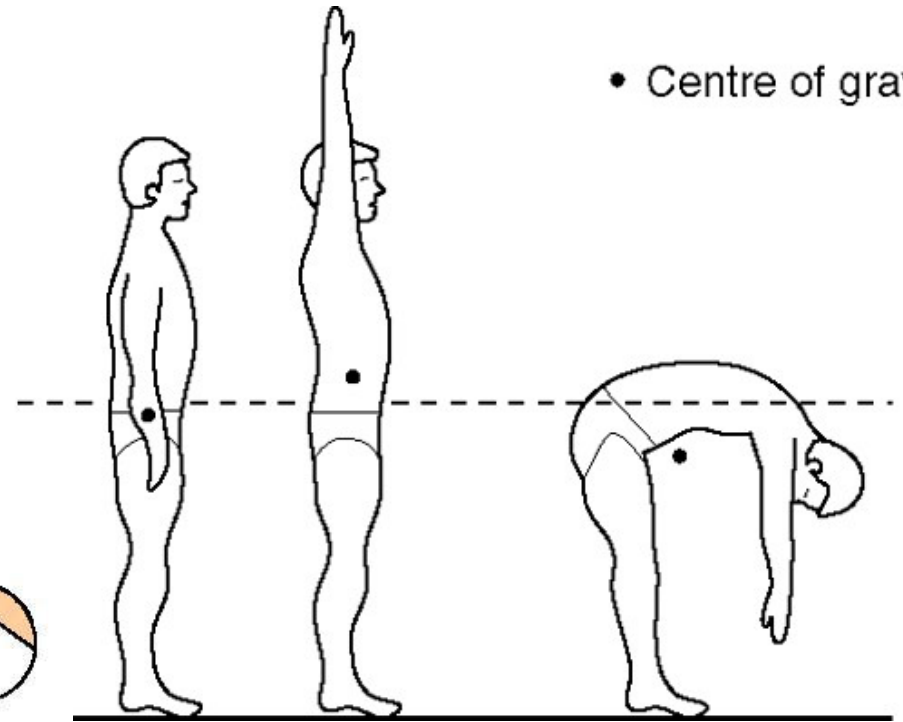
5	Centroid and moment of inertia and mass moment of inertia Centroid: Centroid of lines, plane areas and volumes, Examples related to centroid of composite geometry, Pappus – Guldinus first and second theorems. Moment of inertia of planar cross-sections: Derivation of equation of moment of inertia of standard lamina using first principle, Parallel & perpendicular axes theorems, polar moment of inertia, radius of gyration of areas, section modulus. Examples related to moment of inertia of composite geometry	08	15
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Topics covered in Presentation

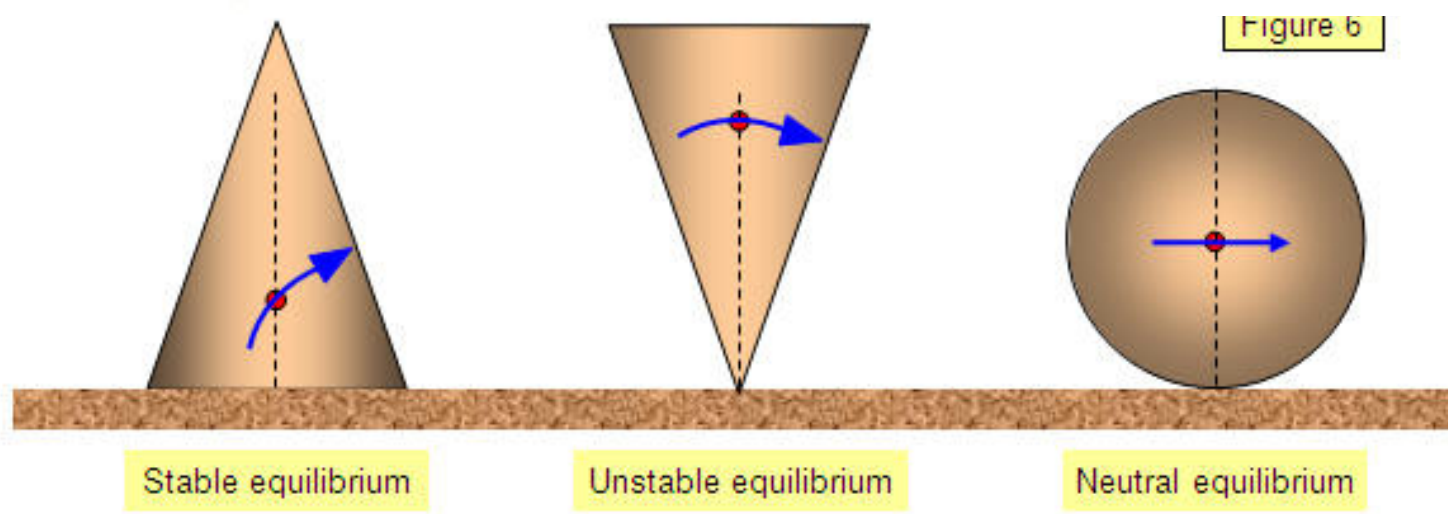
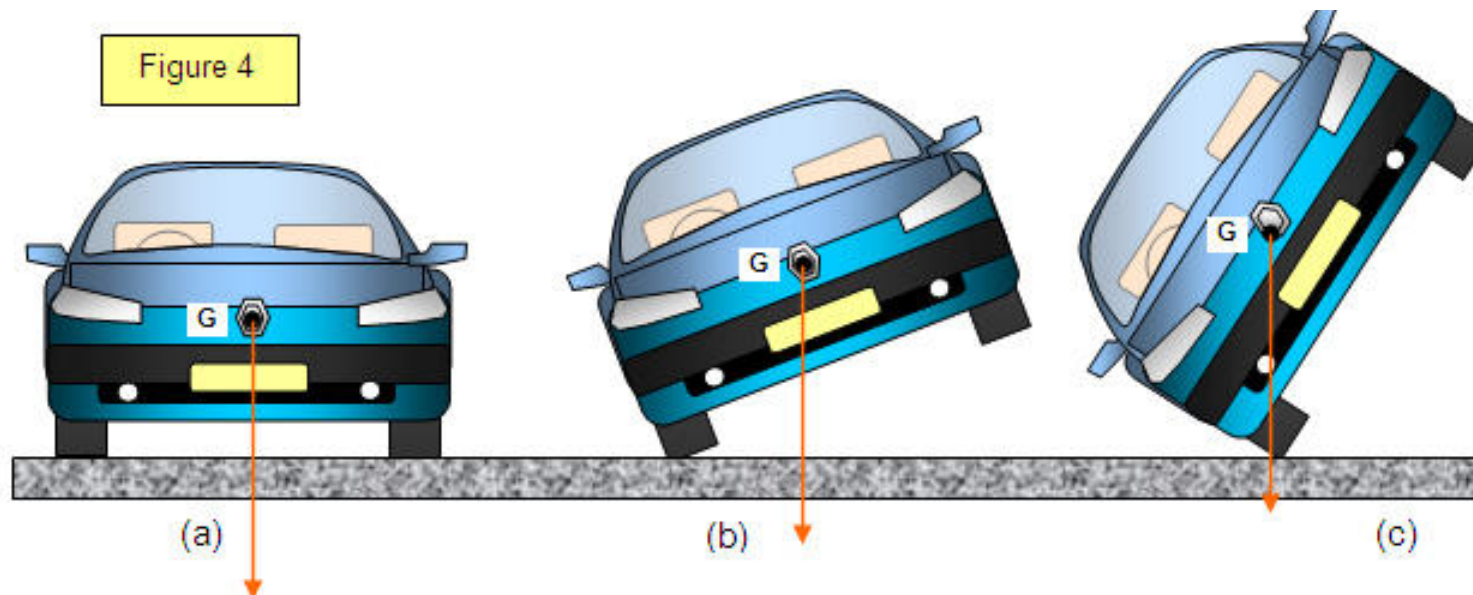
- ▶ Understanding of Centre of gravity
- ▶ Centre of Gravity & Mass
- ▶ Centroid & center of gravity
- ▶ Axis of Reference and Symmetry
- ▶ Types of element
- ▶ Centroid of line segment
- ▶ Line Element Centroid – Basic Shape
- ▶ Examples based on Line element
- ▶ Area Element Centroid – Basic Shape
- ▶ Examples based on Area element or Lamina
- ▶ Pappus Guldinus & its numerical
- ▶ Volume Element Centroid – Basic Shape
- ▶ Examples based on volume elements



• Centre of gravity



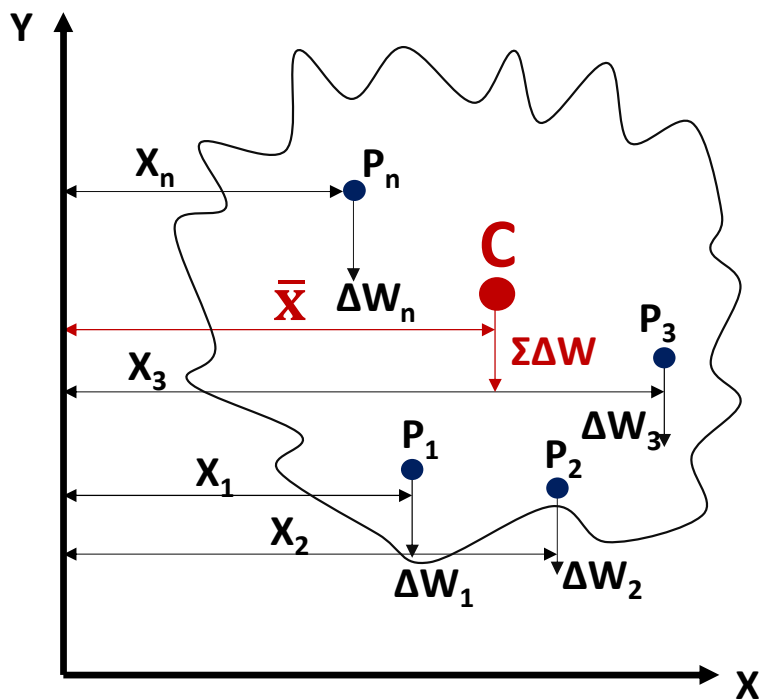






Centre of Gravity

- Centre of Gravity of a body is an imaginary point on which the result of the weight of all constituent particles of the body assumed to be act.



- Let us shown in figure $\Delta W_1, \Delta W_2, \Delta W_3, \dots, \Delta W_n$ are the forces acting on particles of the body due to gravity at distance of $X_1, X_2, X_3, \dots, X_n$ from Y axes and 'C' be the point on which it is assumed that resultant is acting with distance of $\bar{x}$.

- $\Sigma \Delta W = \Delta W_1 + \Delta W_2 + \Delta W_3 + \dots + \Delta W_n$
- Now , applying varignon's theorem, taking moment with respect to Y-axes.
- $\Sigma \Delta W \cdot \bar{x} = \Delta W_1 \cdot X_1 + \Delta W_2 \cdot X_2 + \Delta W_3 \cdot X_3 + \dots + \Delta W_n \cdot X_n$

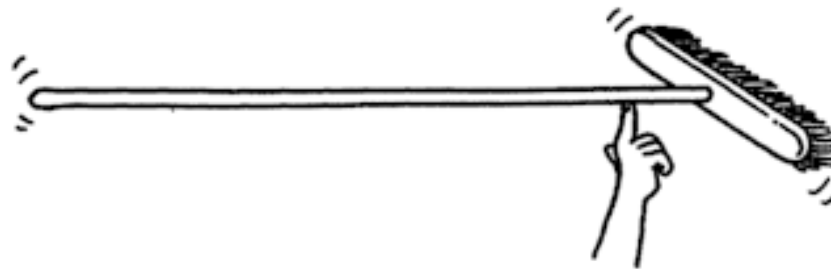
$$\bar{x} = \frac{\Sigma X \cdot \Delta W}{\Sigma \Delta W} \quad \text{Similarly}$$

$$\bar{y} = \frac{\Sigma Y \cdot \Delta W}{\Sigma \Delta W}, \quad \bar{z} = \frac{\Sigma Z \cdot \Delta W}{\Sigma \Delta W}$$

Centre of Mass

► Centre of mass is the point in the body or a system or particles where its mass seems to be concentrated

- Centre of mass concept is used when analysis of bodies under zero gravity is to be carried out e.g. analysis of parts of astronomical vehicles.
- The concept of centre of mass is also very useful in dynamics wherein Newton's second law ($F = m \cdot a$) is frequently used where 'm' is the mass of body and 'a' is its acceleration.



$$\bar{x}_m = \frac{\sum m \cdot x}{\sum m}$$

$$\bar{y}_m = \frac{\sum m \cdot y}{\sum m}$$

$$\bar{z}_m = \frac{\sum m \cdot z}{\sum m}$$

Centroid

- ▶ It is define as geometrical center of the body. OR
- ▶ The point at which the total area of a plane figure (like rectangle, square, triangle, quadrilateral, circle etc.) is assumed to be concentrated is know as the centroid of the area.

Difference between centroid and centre of gravity

Centroid

- It is define as a point about which the entire line, area, volume is assumed to be concentrated.
- It is related to distribution of length, area & volume
- It is used to represent Arc, Rectangle, Square, Triangle, Circle etc.
- Centroid word is used for one dimensional and two dimensional bodies

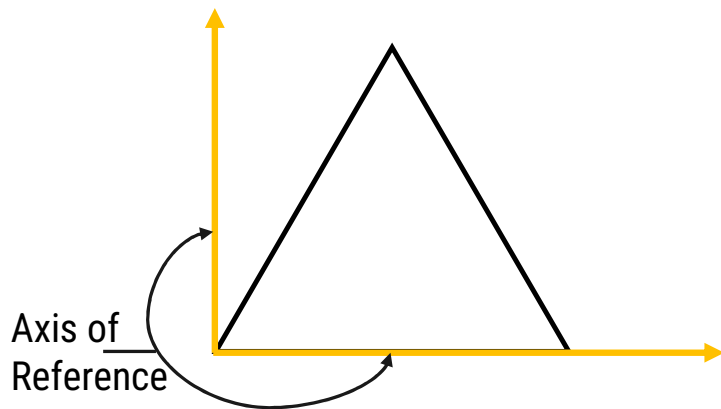
Centre of Gravity

- It is define as a point about which the entire weight of the body is assumed to be concentrated
- It is related to distribution of mass
- It is used to represent Cube, Sphere, cone, cuboid etc.
- C.G. word is used to represent centre of solid bodies or three dimensional bodies

Axis of Reference and Axis of Symmetry

• Axis of Reference

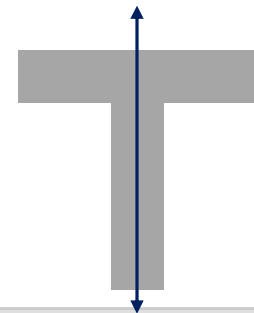
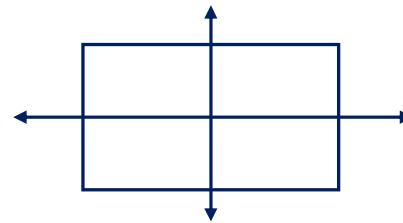
- The C.G. of a body is always calculated with respect to some assumed axis known as axis of reference
- It is generally taken as the lowest line of the figure for calculating $\bar{Y}$ and the left line of figure for calculating $\bar{X}$.



• Axis of Symmetry



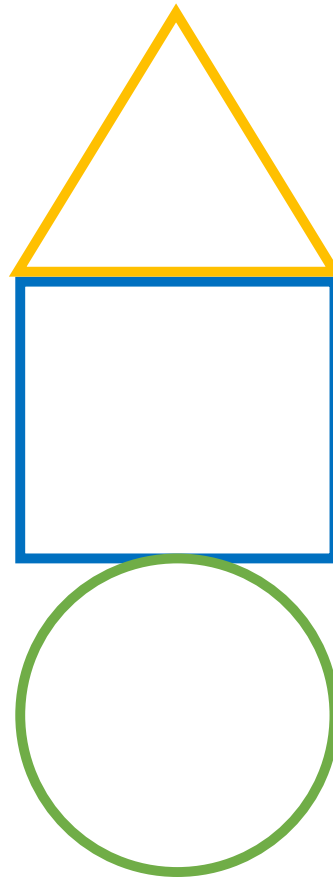
- It is the axis which divide the whole geometry into to equal part either in one direction or in both direction known as the axis of symmetry.
- Before starting to calculate example always check whether the geometry is symmetrical or not?
- If it is symmetrical about any of axis then the C.G. of the body is very much simplified as we have to calculate either $\bar{X}$ or $\bar{Y}$.



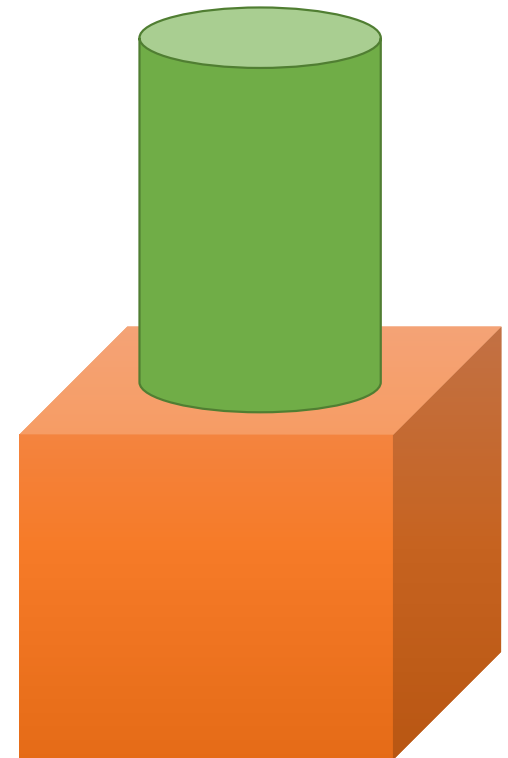
Types of Element



One Dimensional Elements



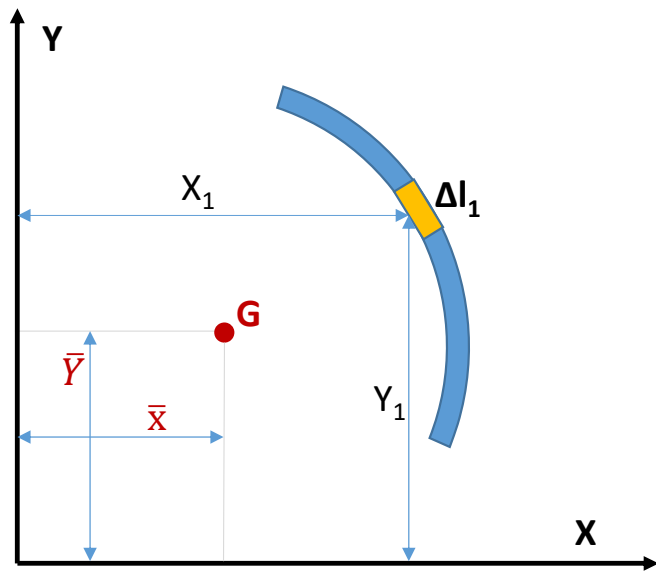
Two Dimensional Elements



Three Dimensional Elements

Centroid of Line Segment

- ▶ Let us consider a line segment of length 'l' which is assumed to be made up of small segments



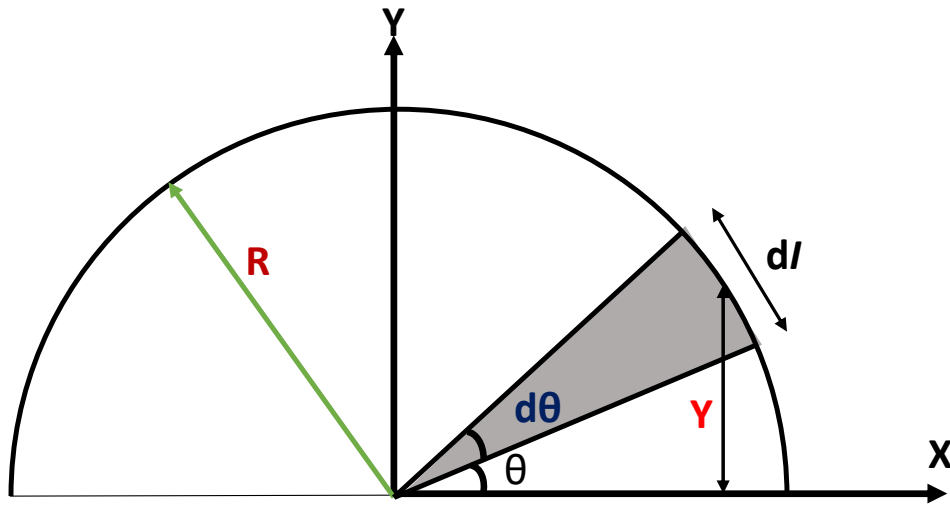
- ▶ Let us consider one segment of length Δl_1 , which is at distance X_1 from Y - axis and ' Y_1 ' from X - axis.
- ▶ Let us assume that centroid of entire element lies at G point which is at $\bar{x}$ and $\bar{y}$ distance from Y axis and X axis respectively.
- ▶ Considering the entire length of line and taking moments of line element about Y axis, we get.

$$l \cdot \bar{x} = \Delta l_1 \cdot X_1 + \Delta l_2 \cdot X_2 + \dots + \Delta l_n \cdot X_n$$

$$\bar{x} = \frac{\sum X_i \cdot \Delta l_i}{\sum l} \quad \text{Or} \quad \bar{x} = \frac{\int X dl}{\int dl}$$

$$\bar{y} = \frac{\sum Y_i \cdot \Delta l_i}{\sum l} \quad \text{Or} \quad \bar{y} = \frac{\int Y dl}{\int dl}$$

Derivation of Equation for Centroid of Semi Circular Arc



- ▶ Consider an elementary length dl at an angular distance ' θ ' from X-axis and subtend an angle of ' $d\theta$ ' at centre 'O' as shown figure.
- ▶ The line is symmetrical w.r.t. Y-axis hence $\bar{x} = 0$ and $\bar{Y}$ is given by following equation

Here, $\frac{y}{R} = \sin \theta$, $\therefore y = R \sin \theta$

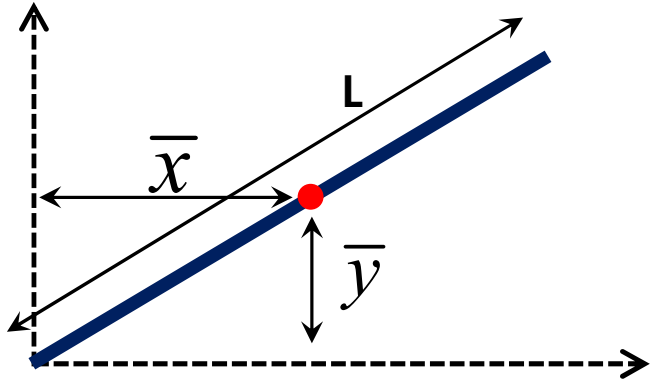
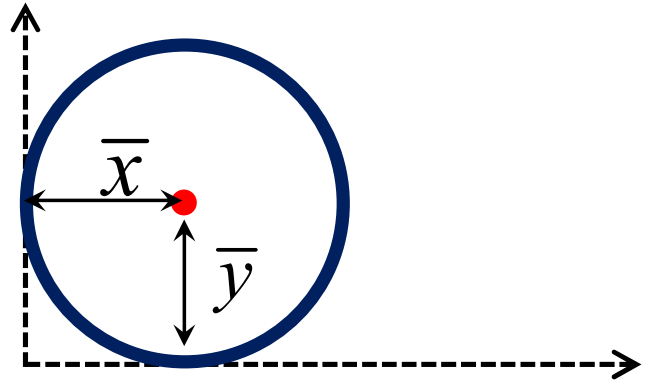
$$\therefore \frac{dl}{R} = d\theta \quad \therefore dl = R \cdot d\theta$$

$$\bar{y} = \frac{\int y \cdot dl}{\int dl} = \frac{\int R \cdot \sin \theta \cdot R \cdot d\theta}{\int R \cdot d\theta}$$

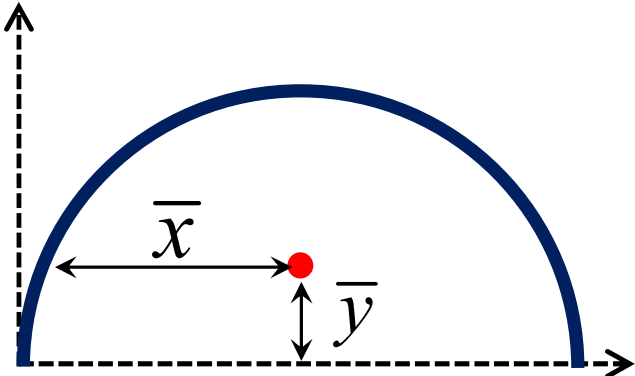
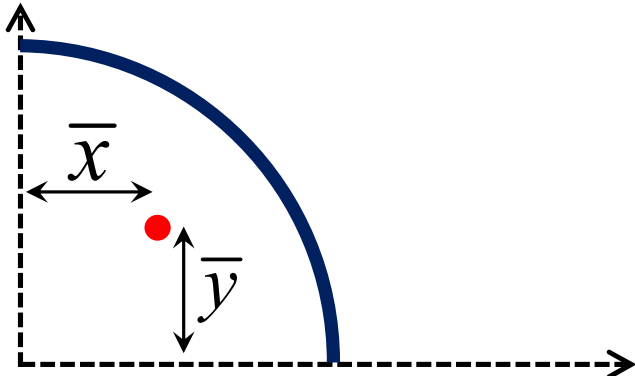
$$\therefore \bar{y} = \frac{\int R^2 \sin \theta \cdot d\theta}{R \int d\theta} = \frac{\int_0^{\pi} \sin \theta \cdot d\theta}{\int_0^{\pi} d\theta}$$

$$\bar{y} = \frac{R \int_0^{\pi/2} \sin \theta \cdot d\theta}{\int_0^{\pi/2} d\theta} = \frac{R[-\cos \theta]_0^{\pi/2}}{\left[\frac{\pi}{2} - 0\right]} \quad \bar{y} = \frac{2R}{\pi}$$

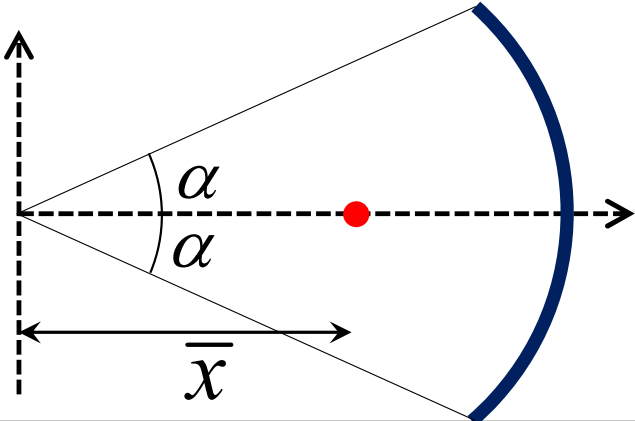
Line Element Centroid – Basic Shape

Element Name	Geometrical Shape	Length	$\bar{x}$	$\bar{y}$
Inclined Line		L	$\frac{L}{2} \cos \theta$	$\frac{L}{2} \sin \theta$
Circular wire		$2\pi r$	r	r

Line Element Centroid – Basic Shape

Element Name	Geometrical Shape	Length	$\bar{x}$	$\bar{y}$
Half circular wire		πr	r	$\frac{2r}{\pi}$
Quarter circular		$\frac{\pi r}{2}$	$\frac{2r}{\pi}$	$\frac{2r}{\pi}$

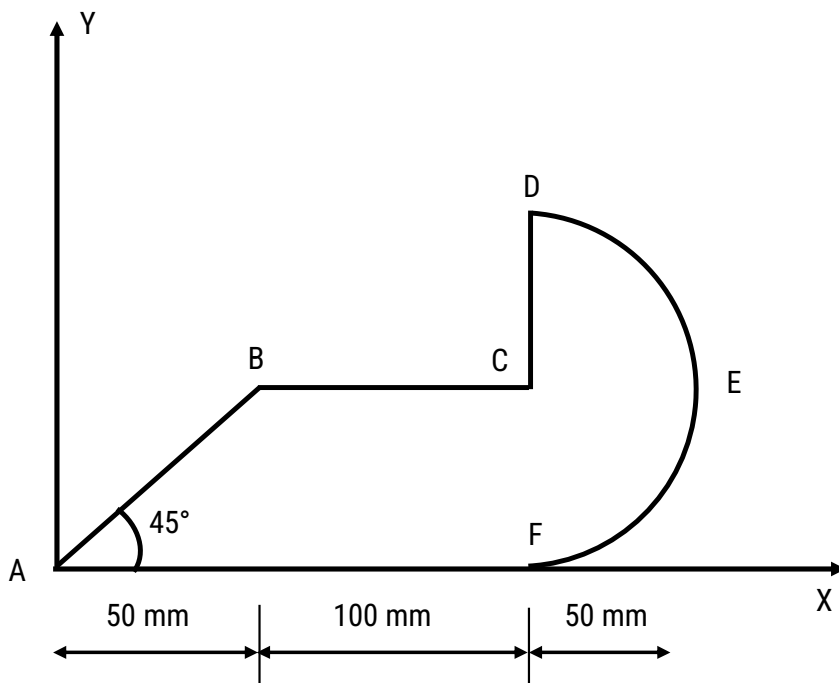
Line Element Centroid – Basic Shape

Element Name	Geometrical Shape	Length	$\bar{x}$	$\bar{y}$
Circular arc		$2r\alpha$ (α in radian)	$\frac{r \sin \alpha}{\alpha}$	On Axis of Symmetry

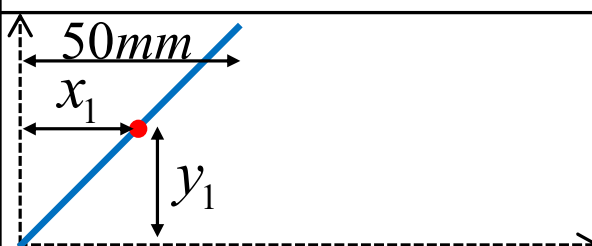
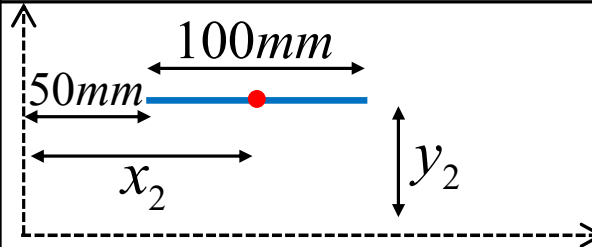
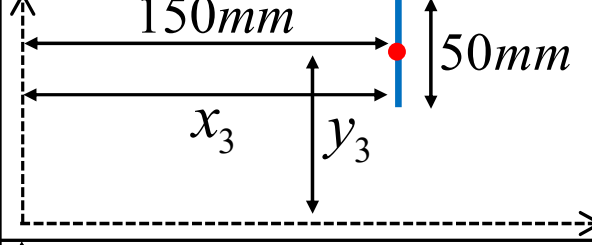
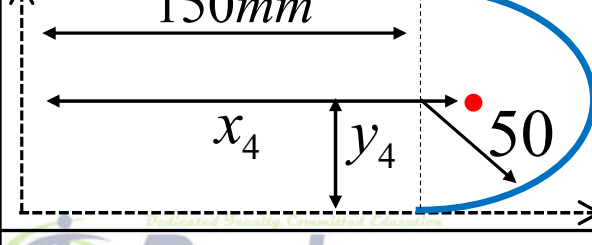
Examples based on Line element

Ex - 1

Determine the centroid of bar bent in to a shape as shown in fig.



- For finding out the centroid of given bar, let's divide the bar in to four numbers of element which is as follows,
 1. Inclined line AB
 2. Straight horizontal line BC
 3. Straight vertical line CD
 4. Semi circular arc DEF

Member	Length (mm)	$x(\text{mm})$	$l \cdot x(\text{mm})^2$	$y(\text{mm})$	$l \cdot y(\text{mm})^2$
	$l_1 = \sqrt{50^2 + 50^2}$ $l_1 = 70.71$	$x_1 = \frac{70.71}{2} \cos 45$ $x_1 = 25$	$l_1 \cdot x_1 = 1767.75$	$y_1 = \frac{70.71}{2} \sin 45$ $y_1 = 25$	$l_1 \cdot y_1 = 1767.75$
	$l_2 = 100$	$x_2 = 50 + \frac{100}{2}$ $x_2 = 100$	$l_2 \cdot x_2 = 10000$	$y_2 = 50$	$l_2 \cdot y_2 = 5000$
	$l_3 = 50$	$x_3 = 50 + 100$ $x_3 = 150$	$l_3 \cdot x_3 = 7500$	$y_3 = 50 + \frac{50}{2}$ $y_3 = 75$	$l_3 \cdot y_3 = 3750$
	$l_4 = \pi \cdot 50$ $l_4 = 157.08$	$x_4 = 150 + \frac{2 \cdot 50}{\pi}$ $x_4 = 181.83$	$l_4 \cdot x_4 = 28491.17$	$y_4 = 50$	$l_4 \cdot y_4 = 7854$
	$\Sigma l = 377.79$	$\Sigma l \cdot x = 47758.92$	$\Sigma l \cdot y = 18371.75$		

$$\bar{x} = \frac{l_1 x_1 + l_2 x_2 + l_3 x_3 + l_4 x_4}{l_1 + l_2 + l_3 + l_4}$$

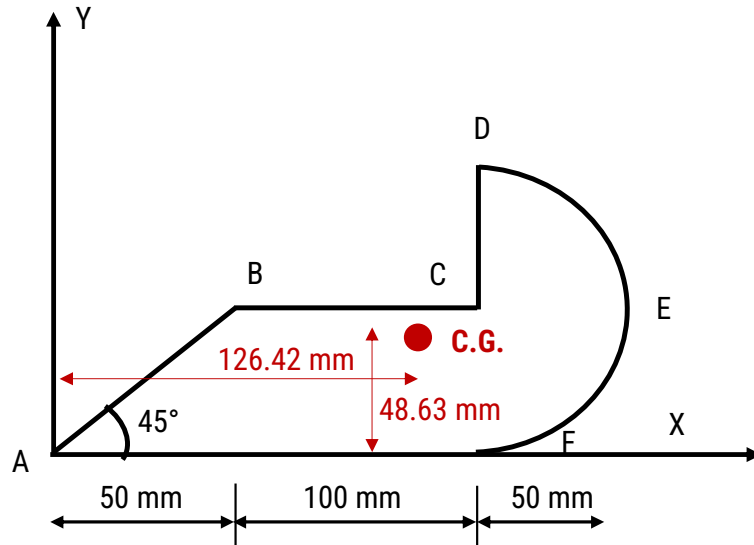
$$\bar{x} = \frac{\Sigma(l \cdot x)}{\Sigma(l)} = \frac{47758.92}{377.79}$$

$$\bar{x} = 126.42 \text{ mm}$$

$$\bar{y} = \frac{l_1 y_1 + l_2 y_2 + l_3 y_3 + l_4 y_4}{l_1 + l_2 + l_3 + l_4}$$

$$\bar{y} = \frac{\Sigma(l \cdot y)}{\Sigma(l)} = \frac{18371.25}{377.79}$$

$$\bar{y} = 48.63 \text{ mm}$$

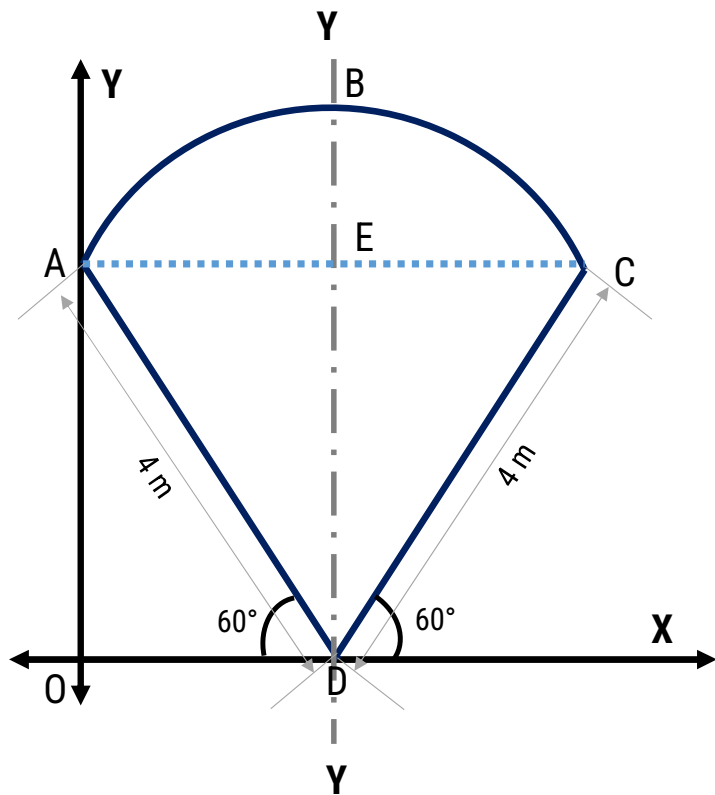


So, Centroid C.G. (126.42 mm, 48.63 mm)

Examples based on Line element

Ex - 2

Determine the centroid of wire bent in to a shape as shown in fig.



- For finding out the centroid of given wire element, let's divide the wire into three numbers of element which is as follows,

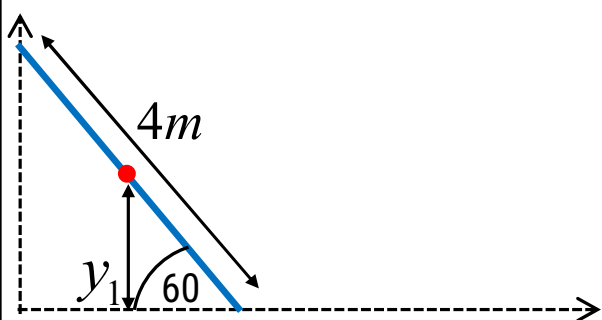
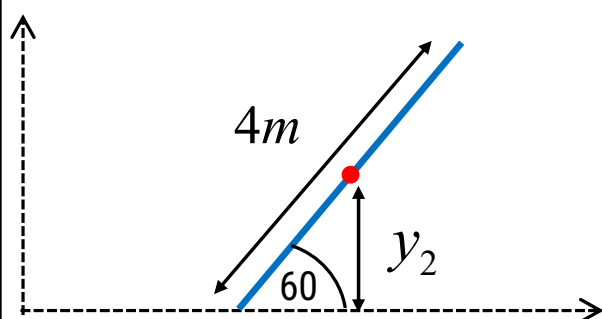
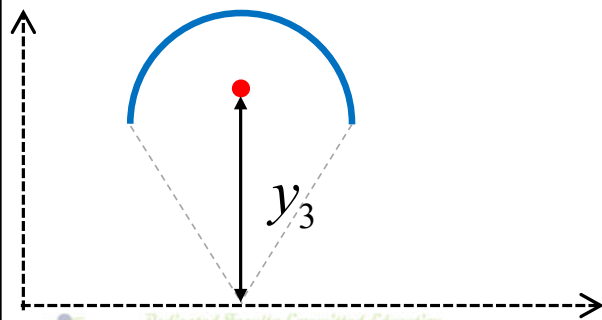
1. Inclined line AD
2. Inclined line CD
3. Semi Circular arc ABC

$$\sin 30 = \frac{AE}{4} \Rightarrow 4 \times \sin 30 = AE = 2m$$

➤ Here given figure is symmetrical to YY axis so that,

$$\bar{x} = AC/2 = 4/2 = 2m$$

$$\bar{x} = 2m$$

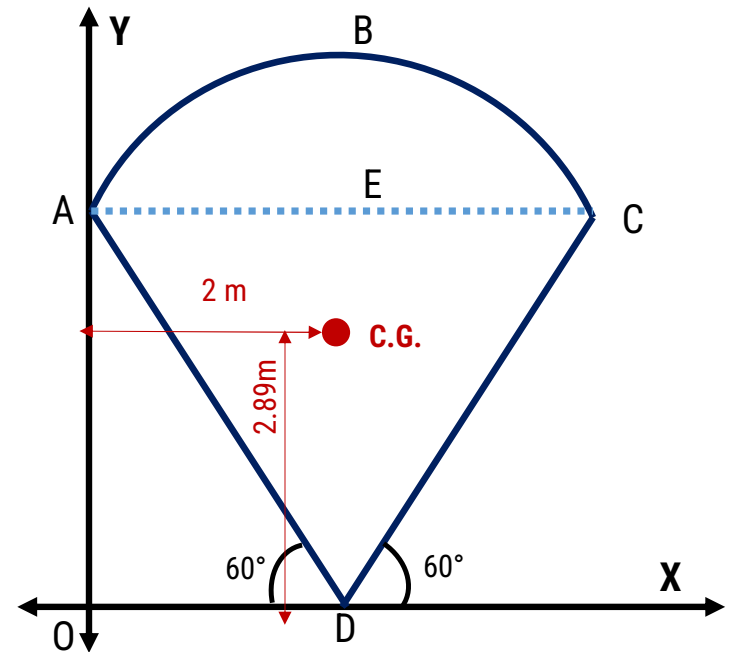
Member	Length (m)	y (m)	$l \cdot y$ (m) ²
	$l_1 = 4$	$y_1 = \frac{4}{2} \sin 60$ $y_1 = 1.732$	$l_1 \cdot y_1$ $= 6.926$
	$l_2 = 4$	$y_2 = \frac{4}{2} \sin 60$ $y_2 = 1.732$	$l_2 \cdot y_2$ $= 6.926$
	$l_3 = \pi \times 2 = 6.28$	$y_3 = 3.46 + \frac{2 \times 2}{\pi}$ $y_3 = 4.374$	$l_3 \cdot y_3$ $= 27.47$
$\Sigma l = 14.28$		$\Sigma l \cdot y = 41.322$	

$$\bar{y} = \frac{l_1 y_1 + l_2 y_2 + l_3 y_3 + l_4 y_4}{l_1 + l_2 + l_3 + l_4}$$

$$\bar{y} = \frac{\sum l_n \times y_n}{\sum l_n}$$

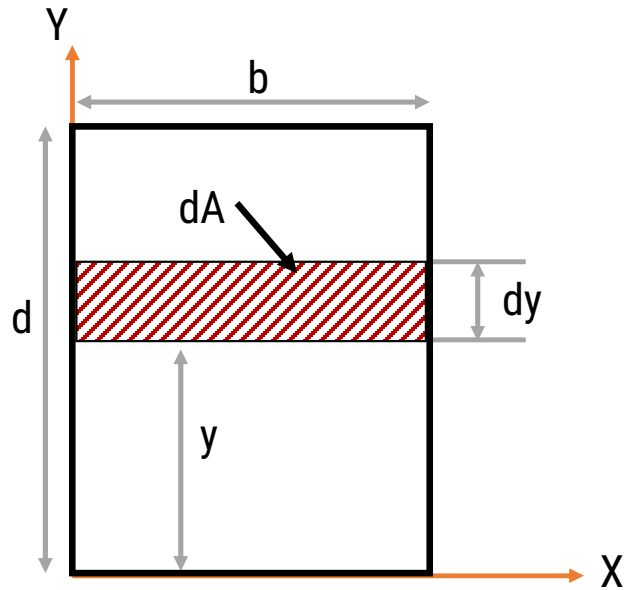
$$\bar{y} = \frac{\sum (l \times y)}{\sum (l)} = \frac{41.322}{14.28}$$

$$\bar{y} = 2.89m$$



So, Centroid C.G. (2 m, 2.89 m)

Centroid of Rectangular Area



- Consider elementary strip 'dA' at a distance Y from X-axis.
- Thickness of strip is 'dy'.
- The position of centroid is given by

$$\bar{y} = \frac{\int y \cdot dA}{\int dA} \quad \text{But, } dA = b \times dy$$

$$\bar{y} = \frac{\int_0^d y \times b \times dy}{\int_0^d b \times dy} = \frac{b \int_0^d y \times dy}{b \int_0^d dy}$$

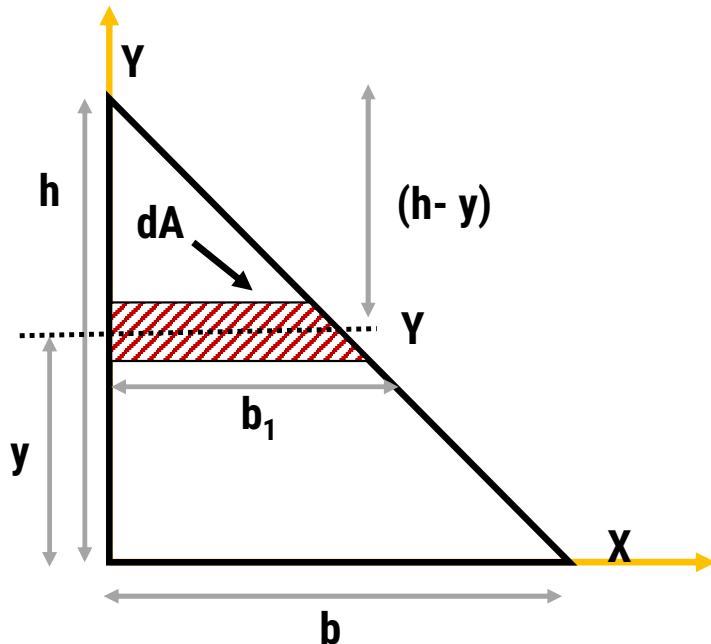
$$\bar{y} = \frac{\left[\frac{y^2}{2} \right]_0^d}{\left[y \right]_0^d} \quad \bar{y} = \frac{d^2}{d}$$

$$\bar{y} = \frac{d}{2}$$

Similarly,

$$\bar{x} = \frac{b}{2}$$

Centroid of a triangle area



Triangular Similarity,

$$\frac{b_1}{b} = \frac{(h-y)}{h}$$

$$b_1 = \frac{b}{h}(h-y)$$

$$dA = b_1 \times dy$$

$$dA = \frac{b}{h}(h-y) \times dy$$

$$\bar{y} = \frac{\int y \times dA}{\int dA}$$

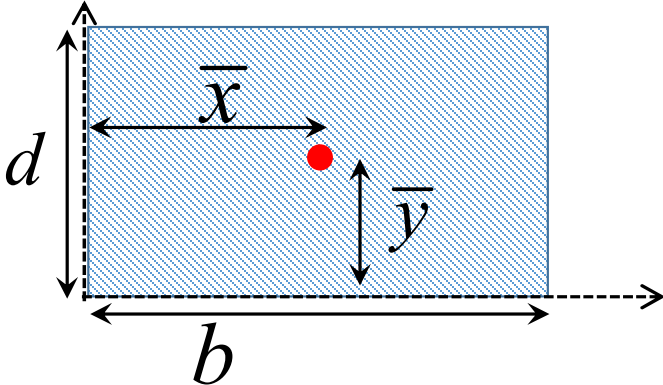
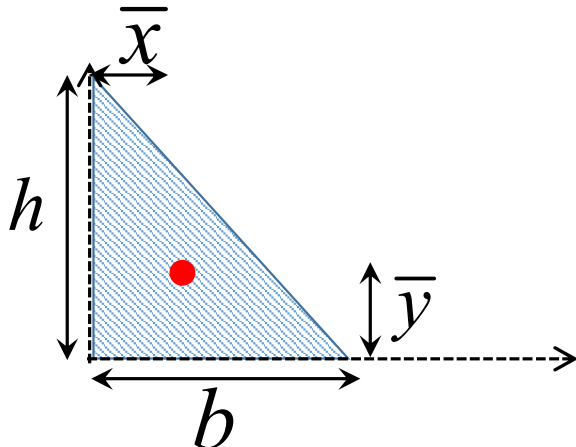
$$\bar{y} = \frac{\int_0^h y \times \frac{b}{h}(h-y) dy}{\int_0^h \frac{b}{h} \times (h-y) dy} \Rightarrow \bar{y} = \frac{\left[\frac{by^2}{2} - \frac{by^3}{3h} \right]_0^h}{\left[by - \frac{by^2}{2h} \right]_0^h}$$

$$\bar{y} = \frac{\frac{bh^2}{2} - \frac{bh^2}{3}}{bh - \frac{bh}{2}}$$

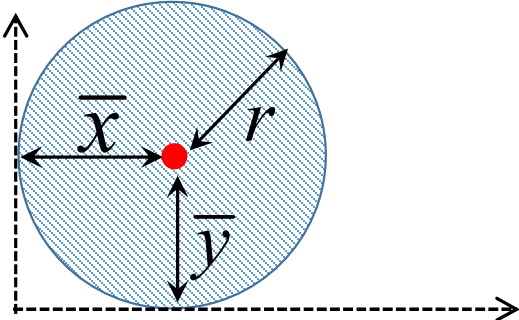
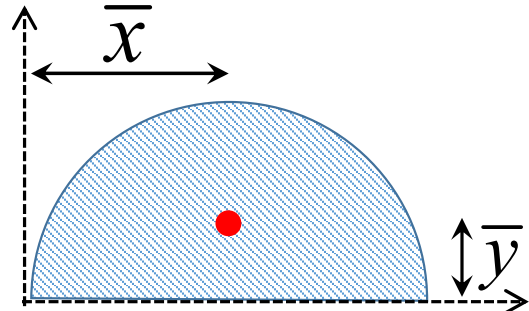
Similarly,

$$\bar{y} = h/3$$

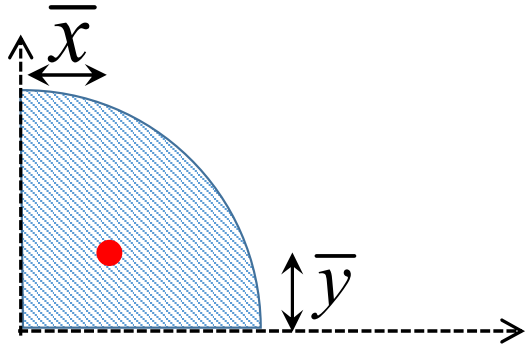
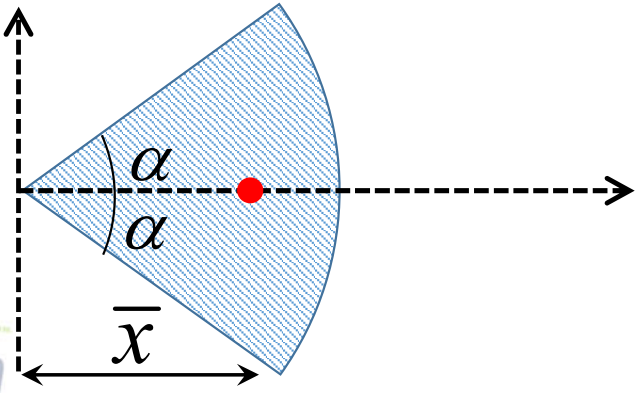
Area Element Centroid – Basic Shape

Element Name	Geometrical Shape	Area	$\bar{x}$	$\bar{y}$
Rectangle		$b \times d$	$\frac{b}{2}$	$\frac{d}{2}$
Triangle		$\frac{1}{2} b \times h$	$\frac{b}{3}$	$\frac{h}{3}$

Area Element Centroid – Basic Shape

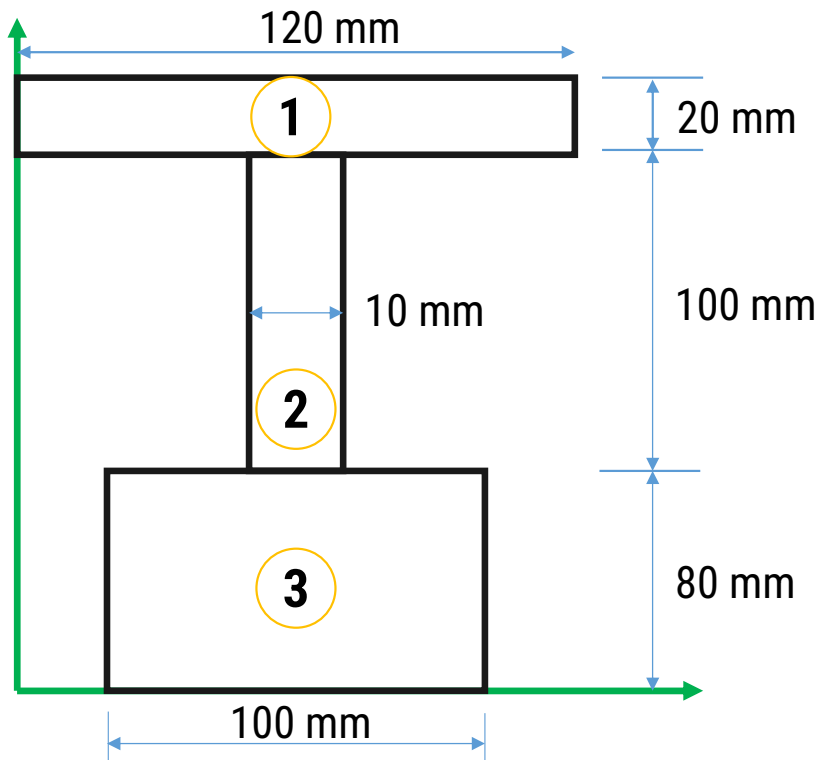
Element Name	Geometrical Shape	Area	$\bar{x}$	$\bar{y}$
Circle		πr^2	r	r
Semicircle		$\frac{\pi r^2}{2}$	r	$\frac{4r}{3\pi}$

Area Element Centroid – Basic Shape

Element Name	Geometrical Shape	Area	$\bar{x}$	$\bar{y}$
Quarter circle		πr^2	$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$
Circular segment		αr^2 (α in radian)	$\frac{2r \sin \alpha}{3\alpha}$	On Axis of Symmetry

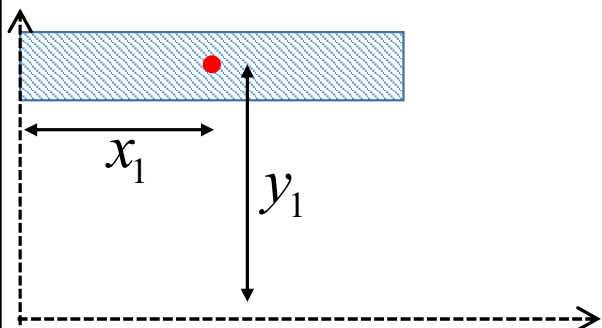
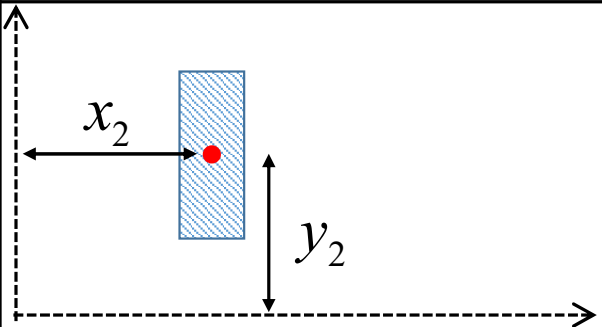
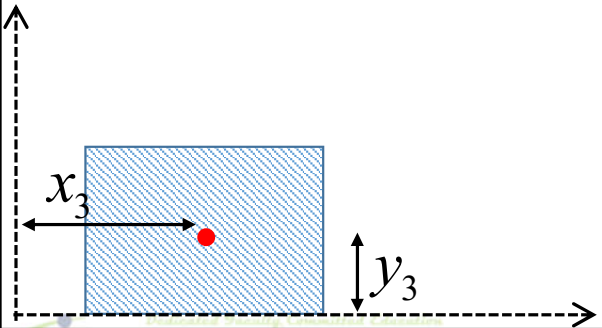
Examples based on Area element or Lamina

Ex - 3 Determine the centroid of shape as shown in fig.



Given geometry can be divide into 3 part,

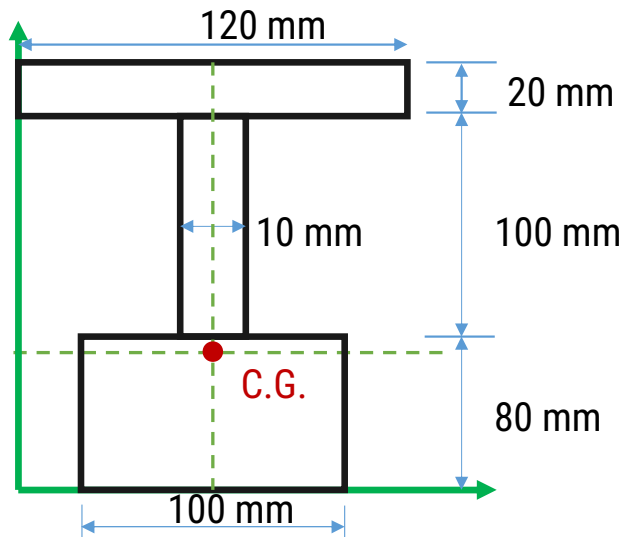
1. Upper horizontal Rectangle
2. Vertical Rectangle
3. Lower Horizontal Rectangle

Member	Area (mm) ²	x(mm)	A · x(mm) ³	y (mm)	A · y(mm) ³
	$A_1 = 120 \times 20$ $= 2400$	$x_1 = \frac{120}{2}$ $= 60$	$= 144000$	$y_1 = 80 + 100 + \frac{20}{2}$ $= 190$	$= 456000$
	$A_2 = 10 \times 100$ $= 1000$	$x_2 = 60$	$= 60000$	$y_2 = 80 + \frac{100}{2}$ $= 130$	$= 130000$
	$A_3 = 100 \times 80$ $= 8000$	$x_3 = 60$	$= 480000$	$y_3 = \frac{80}{2}$ $= 40$	$= 320000$
	$\Sigma A = 11400$	$\Sigma A \cdot x = 684000$	$\Sigma A \cdot y = 906000$		

$$\bar{x} = \frac{A_1x_1 + A_2x_2 + A_3x_3}{A_1 + A_2 + A_3}$$

$$\bar{x} = \frac{\Sigma(A \cdot x)}{\Sigma(A)} = \frac{684000}{11400}$$

$$\bar{x} = 60\text{mm}$$



$$\bar{y} = \frac{A_1y_1 + A_2y_2 + A_3y_3}{A_1 + A_2 + A_3}$$

$$\bar{y} = \frac{\Sigma(A \cdot y)}{\Sigma(A)} = \frac{906000}{11400}$$

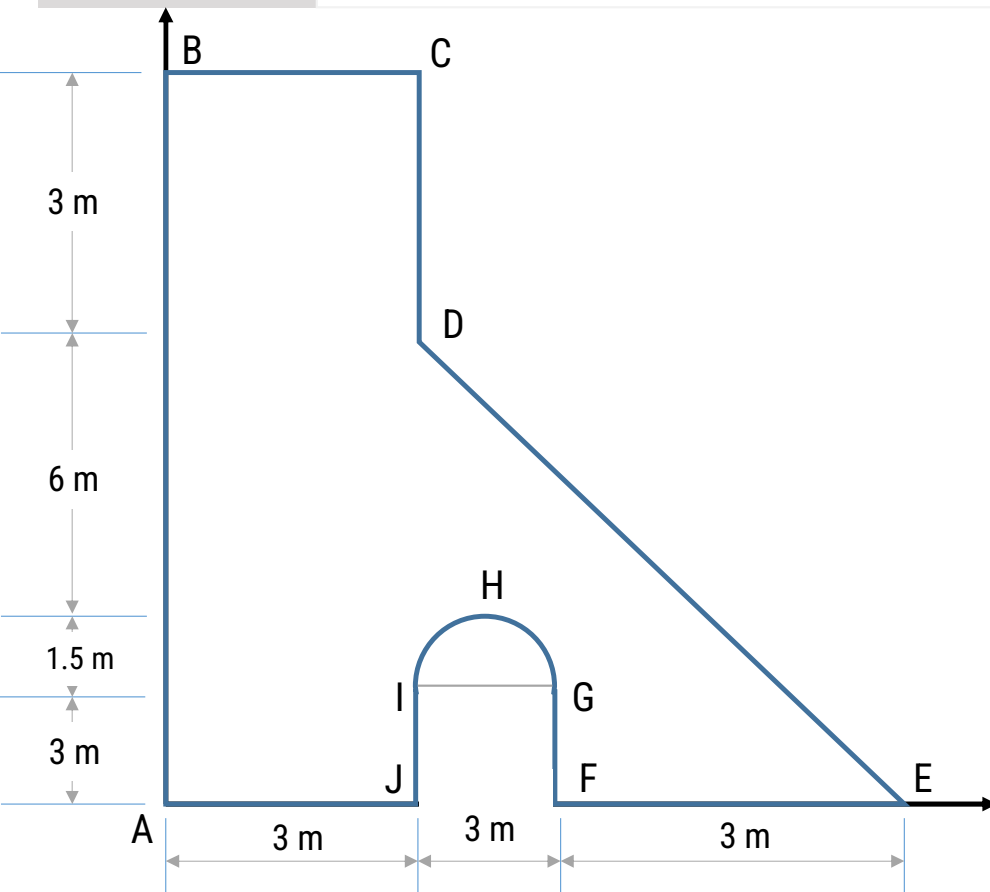
$$\bar{y} = 79.47\text{mm}$$

So, Centroid C.G. (60 mm, 79.47 mm)

Examples based on Area element or Lamina

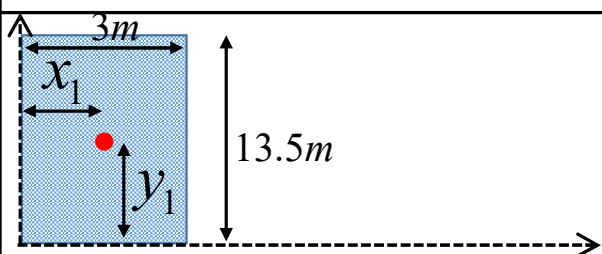
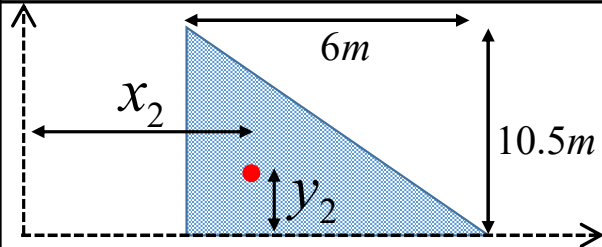
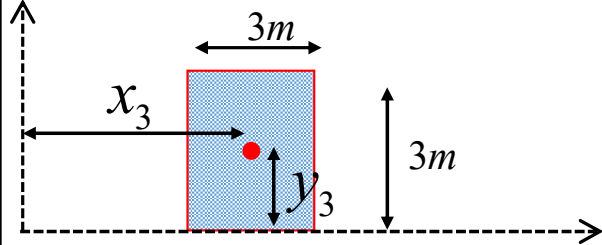
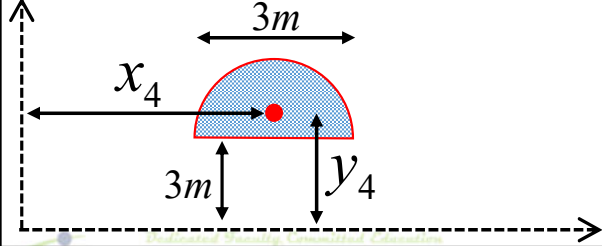
Ex - 4

Determine co-ordinate of centroid with respect to 'A' of the section shown in fig



Given geometry can be divide into 4 part,

- 1) Rectangle ABCJ
- 2) Triangle DJE
- 3) Small Rectangle IGFJ
- 4) Semicircle IHG

Member	Area (m) ²	x (m)	A · x (m) ³	y (m)	A · y (m) ³
	3×13.5 $= 40.5$	$3 / 2$ $= 1.5$	$= 60.75$	$13.5 / 2$ $= 6.75$	$= 273.375$
	$\frac{1}{2} \times 6 \times 10.5$ $= 31.5$	$3 + \frac{1}{3} \times 6$ $= 5$	$= 157.5$	$\frac{1}{3} \times 10.5$ $= 3.5$	$= 110.25$
	-3×3 $= -9$	$3 + \frac{3}{2}$ $= 4.5$	$= -40.5$	$3 / 2$ $= 1.5$	$= -13.5$
	$-\frac{\pi \times 1.5^2}{2}$ $= -3.53$	$3 + \frac{3}{2}$ $= 4.5$	$= -15.89$	$3 + \frac{4 \times 1.5}{3\pi}$ $= 3.64$	$= -12.85$
	$\Sigma A = 59.47$	$\Sigma A \cdot x = 161.86$	$\Sigma A \cdot y = 357.275$		

$$\bar{x} = \frac{A_1x_1 + A_2x_2 - A_3x_3 - A_4x_4}{A_1 + A_2 - A_3 - A_4} \quad \bar{y} = \frac{A_1y_1 + A_2y_2 - A_3y_3 - A_4y_4}{A_1 + A_2 - A_3 - A_4}$$

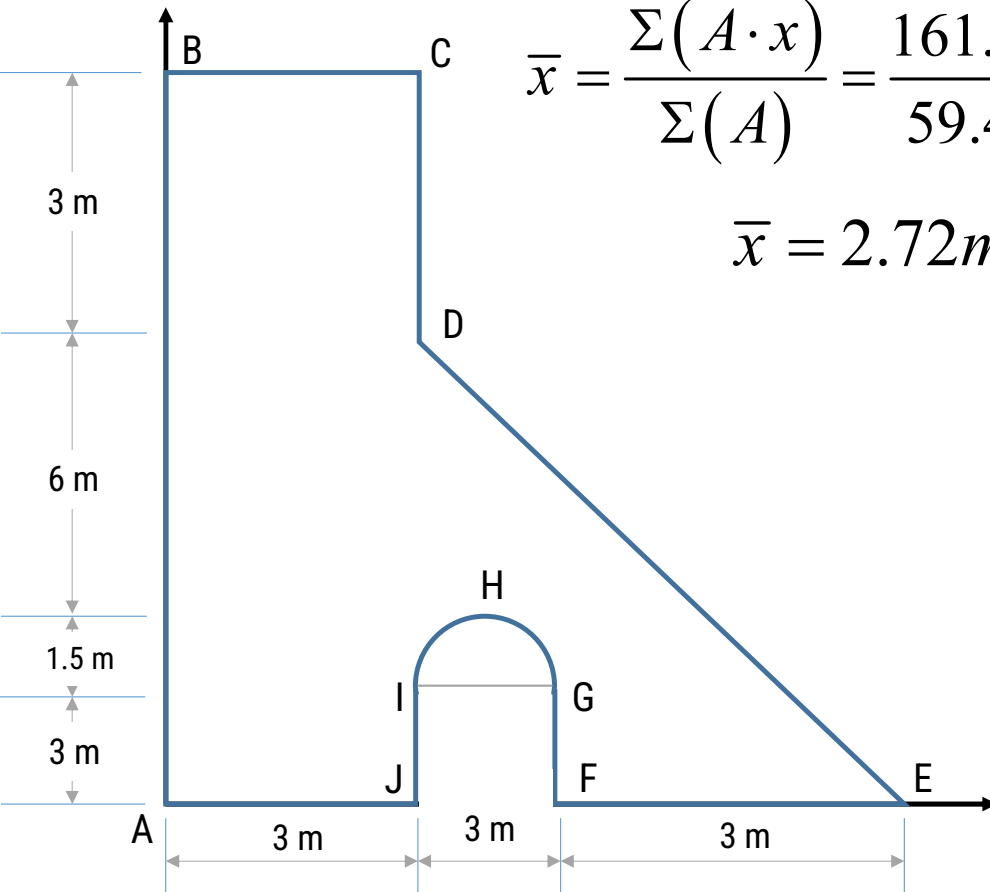
$$\bar{x} = \frac{\Sigma(A \cdot x)}{\Sigma(A)} = \frac{161.86}{59.47}$$

$$\bar{x} = 2.72m$$

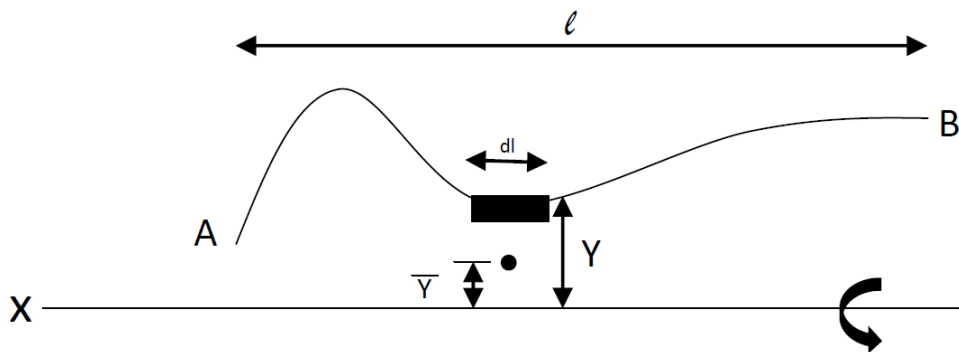
$$\bar{y} = \frac{\Sigma(A \cdot y)}{\Sigma(A)} = \frac{357.275}{59.47}$$

$$\bar{y} = 6m$$

So, Centroid C.G. (2.72 m, 6 m)



Pappus Guldinus First Theorem



➤ This theorem states that, “The area of surface of revolution is equal to the product of length of generating curves & the distance travelled by the centroid of the generating curve while the surface is being generated”.

➤ As shown in fig. consider small element having length dl & at ‘ y ’ distance from $x - x$ axis.

➤ Surface area dA by revolving this element $dA = 2\pi y \cdot dl$ (complete revolution)

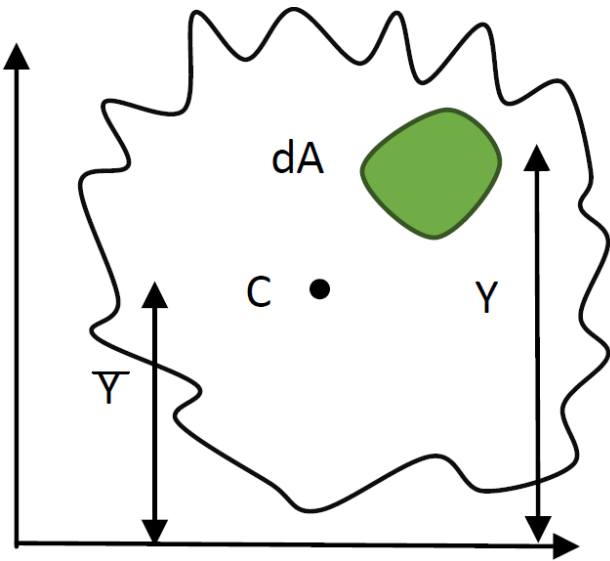
➤ Now, total area,
$$A = \int dA = \int 2\pi \cdot y dl = 2\pi \int y \cdot dl$$

$$A = 2\pi \cdot \bar{y} \cdot l$$

➤ When the curve rotate by an angle ‘ θ ’

$$A = 2\pi \cdot \bar{y} \cdot l \cdot \frac{\theta}{2\pi} = \theta \cdot \bar{y} \cdot l$$

Pappus Guldinus Second Theorem



- ▶ This theorem states that, "The volume of a body of revolution is equal to the product of the generating area & distance travelled by the centroid of revolving area while rotating around its axis of rotation"
- ▶ Consider area 'dA' as shown in fig. the volume generated by revolution will be

$$dV = 2 \cdot \pi \cdot y \cdot dA$$

- ▶ Now, the total volume generated by lamina,

$$V = \int dV = \int 2\pi \bar{y} dA = 2\pi \bar{y} A \text{ (completed revolution)}$$

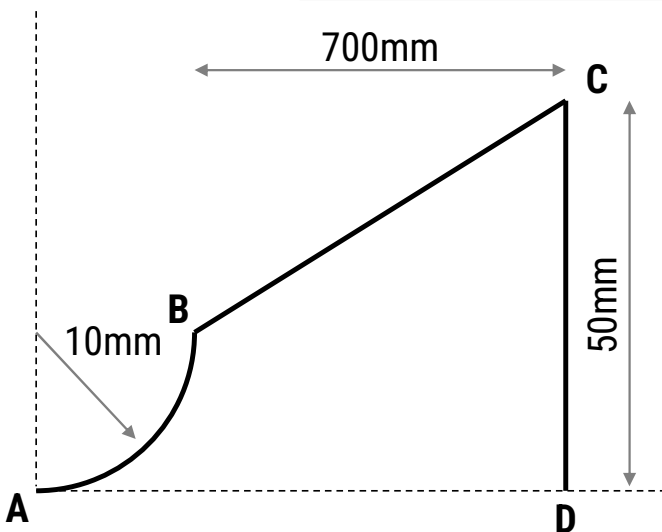
- ▶ When the area revolves about ' θ ' angle volume will be

$$V = 2\pi \bar{y} A \frac{\theta}{2\pi} = \theta \cdot \bar{y} \cdot A$$

Examples based on Pappus Guldinus First Theorem

Ex - 5

Determine surface area of revolution of the length ABCD revolved about x-axis



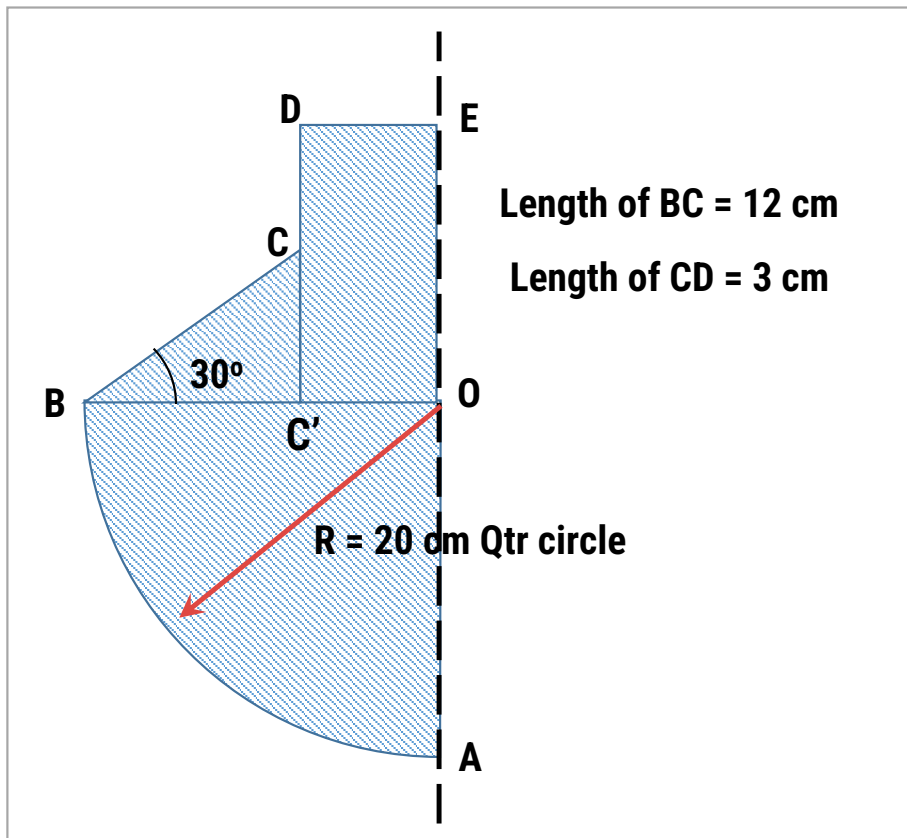
► Surface of revolution about AD

Part	Length (l) mm	Centroidal Distance $\bar{y}$ -bar (mm)	Surface area $2\pi \bar{y}l$ (mm^2)
AB	$\frac{\pi r}{2} = \frac{\pi \times 10}{2} = 15.7$	$10 - \frac{2 \times 10}{\pi} = 3.6$	355
BC	$\sqrt{700^2 + 40^2} = 701.1$	$10 + \frac{40}{2} = 30$	132154
CD	50	$(50 / 2) = 25$	7854
			$\Sigma 2\pi \bar{y}l = 140363 \text{mm}^2$

Surface area of revolution = 140363 mm^2

Examples based on Pappus Guldinus Second Theorem

Ex - 6 Determine volume of revolution generated by revolving plane lamina ABCDEA shown in fig. about y – y axis



► From geometry

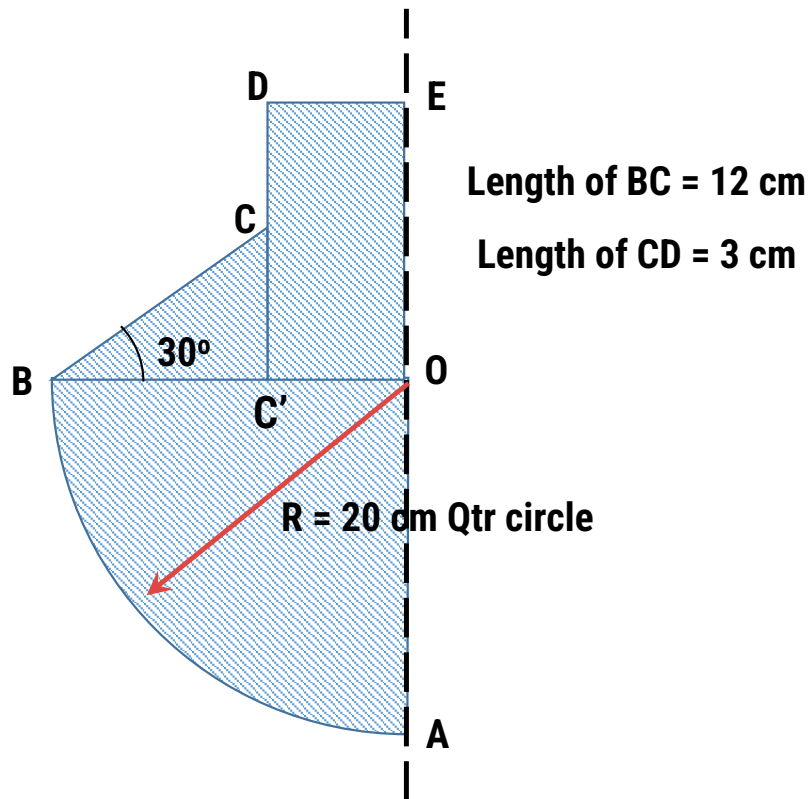
$$BC = 12 \text{ cm} \quad CD = 3 \text{ cm}$$

$$CC' = BC \sin 30 = 12 \sin 30 = 6 \text{ cm}$$

$$BC' = 12 \cos 30 = 10.392 \text{ cm}$$

$$DE = 20 - 10.392 = 9.608 \text{ cm}$$

$$DC' = DC + CC' = 3 + 6 = 9 \text{ cm}$$



$$BC = 12\text{ cm} \quad CD = 3\text{ cm} \quad CC' = 6\text{ cm}$$

$$BC' = 10.392\text{ cm} \quad DE = 9.608\text{ cm} \quad DC' = 9\text{ cm}$$

Item	Area (cm ²)	X bar (cm)
Rectangle	$9.608 \times 9 = 86.472$	$9.608 / 2 = 4.804$
Triangle	$(1/2)10.392 \times 6 = 31.176$	$9.608 + (1/3)10.392 = 13.07$
Quarter circle	$\frac{\pi \times 20^2}{4} = 314.15$	$\frac{4 \times 20}{3\pi} = 8.488$

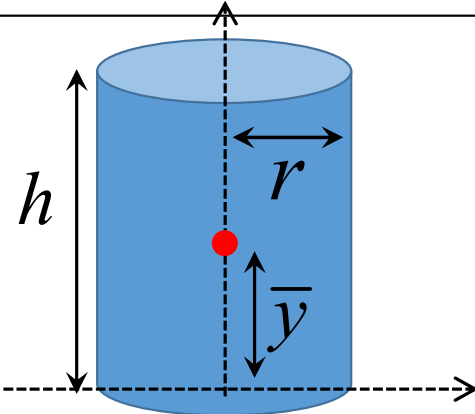
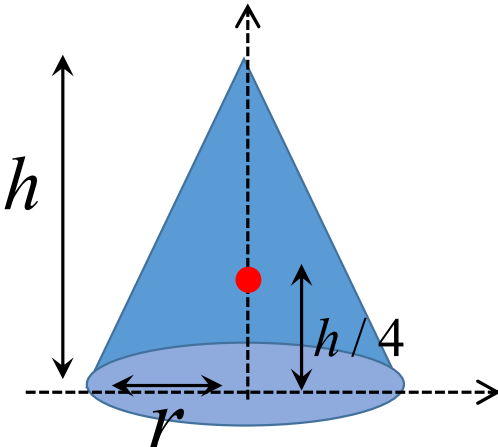
$$\sum A = 431.798\text{ cm}^2$$

► Centroid of given lamina

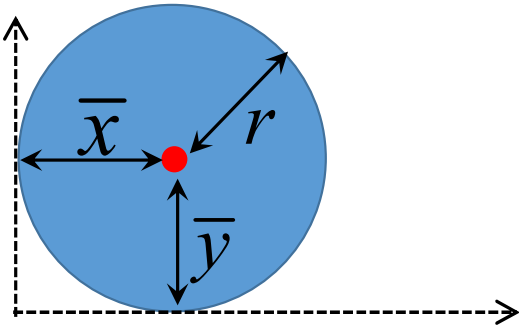
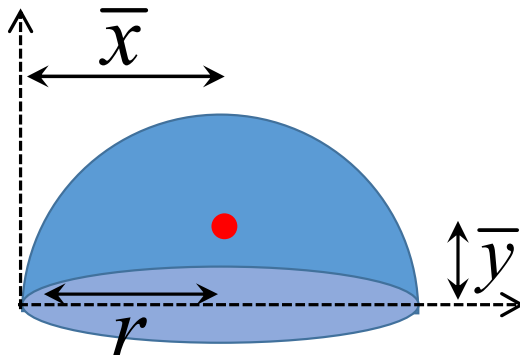
$$\bar{x} = \frac{a_1x_1 + a_2x_2 + a_3x_3}{a_1 + a_2 + a_3} = \frac{86.472 \times 4.804 + 31.176 \times 13.07 + 314.15 \times 8.488}{431.798} = 8.08\text{ cm}$$

$$V = 2\pi\bar{x}A = 2\pi \times 8.08 \times 431.798 = 21921.58\text{ cm}^3$$

Volume Element Centroid – Basic Shape

Element Name	Geometrical Shape	Volume	$\bar{x}$	$\bar{y}$
Solid cylinder	 A diagram of a solid cylinder with height h and radius r. The centroid is marked with a red dot at a distance $\bar{y}$ from the base.	$\pi r^2 h$	0	$\frac{h}{2}$
Solid right Circular cone	 A diagram of a solid right circular cone with height h and radius r. The centroid is marked with a red dot at a distance $h/4$ from the base.	$\frac{1}{3} \pi r^2 h$	0	$\frac{h}{4}$

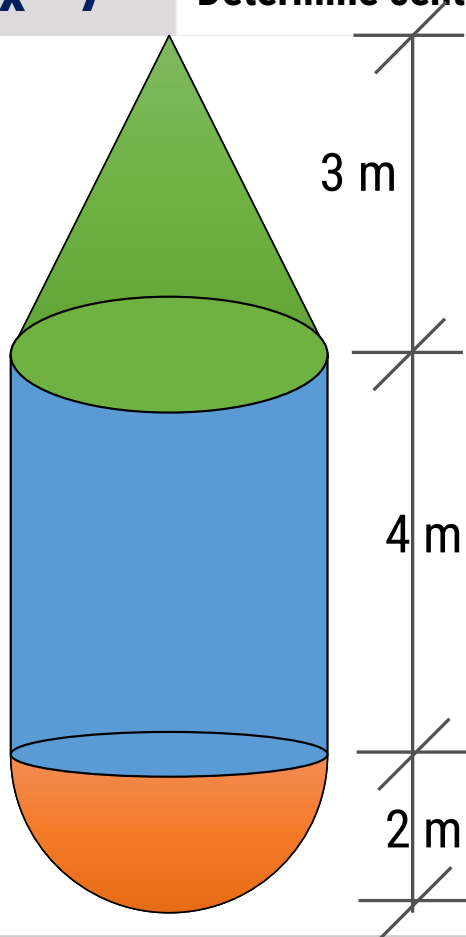
Volume Element Centroid – Basic Shape

Element Name	Geometrical Shape	Volume	$\bar{x}$	$\bar{y}$
Sphere		$\frac{4}{3}\pi r^3$	r	r
Hemisphere		$\frac{2\pi r^3}{3}$	r	$\frac{3r}{8}$

Examples based on Volume element

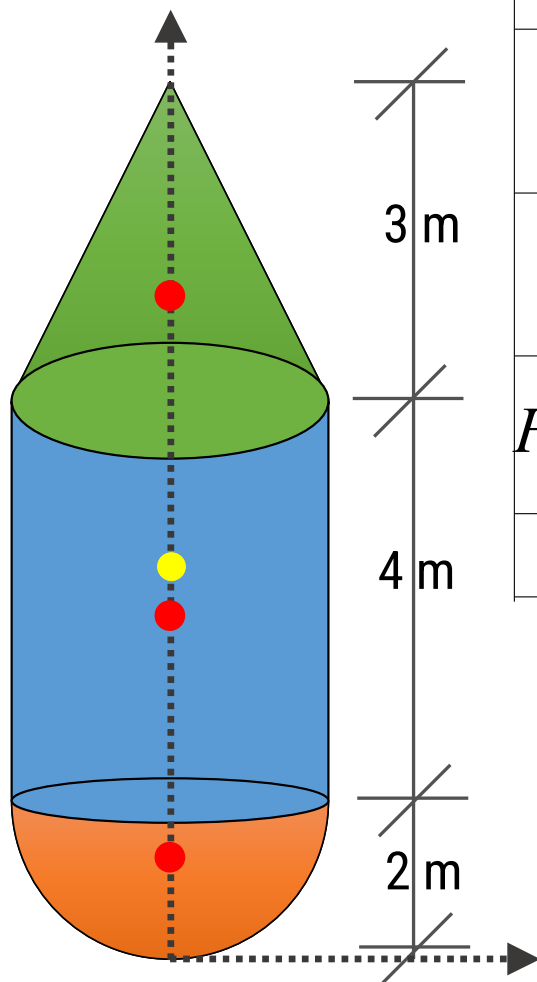
Ex - 7

Determine centroid of body as shown in fig.



Given geometry can be divide into 3 part,

- 1) Top Solid right Circular cone
- 2) Solid cylinder
- 3) Hemisphere



<i>Part</i>	<i>Volume (m)³</i>	<i>y (m)</i>	<i>V · y (m)⁴</i>
<i>Cone</i>	$\frac{\pi \times 2^2 \times 3}{3} = 12.57$	$2 + 4 + \frac{3}{4} = 6.75$	84.85
<i>Cylinder</i>	$\pi \times 2^2 \times 4 = 50.265$	$2 + \frac{4}{2} = 4$	201.06
<i>Hemisphere</i>	$\frac{2\pi \times 2^3}{3} = 16.755$	$2 - \frac{3 \times 2}{8} = 1.25$	20.944
$\Sigma V = 79.59$		$\Sigma V \cdot y = 306.85$	

$$\bar{y} = \frac{V_1 y_1 + V_2 y_2 + V_3 y_3}{V_1 + V_2 + V_3} = \frac{\Sigma(V \cdot y)}{\Sigma(V)}$$

$$\bar{y} = \frac{306.85}{79.59} = 3.855m$$

So, Centroid C.G. (0 m, 3.855 m)